

Exercise 5: Optimization Landscape of Neural Networks

Lecturer: Aurelien Lucchi

Teaching Assistants: Enea Monzio Compagnoni, Jim Zhao

Problem 1 (Matrix Completion):

We consider the problem of matrix sensing where we have a model parametrized by $\mathbf{W} \in \mathbb{R}^{d \times d}$. We observe a set of linear measurements of the form $\langle \mathbf{W}, \mathbf{A}_n \rangle_F$ where $[\mathbf{A}_n]_{ij} \sim \mathcal{N}(0, 1)$. We will further assume that the data is labeled by a matrix sensing model parameterized by $\mathbf{W}^* \in \mathbb{R}^{d \times d}$ (this is sometimes called "planted model" in the literature).

We will study the dynamics of this model trained with gradient flow on a squared loss. As we will soon see, this setting is related to the problem of training a deep linear network. In order to simulate depth, we will consider the matrix $\mathbf{W}$ as the product of a set of square matrices $\mathbf{W}_i \in \mathbb{R}^{d \times d}$, i.e. $\mathbf{W} = \mathbf{W}_L \dots \mathbf{W}_1$.

The objective function is given by

$$\mathcal{L}(\mathbf{W}) = \frac{1}{2L} \mathbb{E}_{\mathbf{A}} \langle \mathbf{W} - \mathbf{W}^*, \mathbf{A} \rangle_F^2 = \frac{1}{2L} \|\mathbf{W}_L \dots \mathbf{W}_1 - \mathbf{W}^*\|_F^2.$$

We will use gradient flow to optimize the parameters and denote by $\mathbf{W}_k(t)$ the matrices $\mathbf{W}_k$ at time t .

a) Denote $\mathbf{W}_{j:k}^\top = \prod_{i=j}^k \mathbf{W}_i^\top = \mathbf{W}_j^\top \mathbf{W}_{j+1}^\top \dots \mathbf{W}_k^\top$. The partial gradient of $\mathcal{L}$ with respect to $\mathbf{W}_k$ (where $k = 1, \dots, L$) is

$$\frac{\partial \mathcal{L}(\mathbf{W})}{\partial \mathbf{W}_k} = \frac{1}{L} \mathbf{W}_{k+1:L}^\top (\mathbf{W} - \mathbf{W}^*) \mathbf{W}_{1:k-1}^\top.$$

What is the gradient flow equation for the matrix $\mathbf{W}_k$?

b) Prove that for all $t \geq 0$ and $k = 1, \dots, L-1$:

$$\mathbf{W}_{k+1}^\top(t) \dot{\mathbf{W}}_{k+1}(t) = \dot{\mathbf{W}}_k(t) \mathbf{W}_k^\top(t). \quad (1)$$

And hence

$$\mathbf{W}_{k+1}^\top(t) \mathbf{W}_{k+1}(t) = \mathbf{W}_k(t) \mathbf{W}_k^\top(t). \quad (2)$$

c) For any $t \geq 0$ and $k = 1, \dots, L$, assume the singular values of $\mathbf{W}_k$ are all distinct and indexed in strictly decreasing order: $\sigma_1 > \dots > \sigma_d > 0$, and let $\mathbf{W}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$ be its (unique) singular value decomposition (SVD). Show that

$$\mathbf{\Sigma}_{k+1} = \mathbf{\Sigma}_k, \quad \mathbf{U}_k = \mathbf{V}_{k+1},$$

for any $t \geq 0$ and $k = 1, \dots, L$.

d) Denote $\mathbf{\Sigma} = \mathbf{\Sigma}_1 = \dots = \mathbf{\Sigma}_L \in \mathbb{R}^{d \times d}$. Now prove that the gradient flow equation can be written as:

$$\dot{\mathbf{W}}_k = \frac{1}{L} \mathbf{V}_{k+1} \mathbf{\Sigma}^{L-k} \mathbf{U}_L^\top \left(\mathbf{W}^* - \mathbf{U}_L \mathbf{\Sigma}^L \mathbf{V}_1^\top \right) \mathbf{V}_1 \mathbf{\Sigma}^{k-1} \mathbf{V}_{k-1}^\top,$$

for all $k = 1, \dots, L$.

e) By the product rule $\dot{\mathbf{W}} = \sum_{k=1}^L \mathbf{W}_{L:k+1} \dot{\mathbf{W}}_k \mathbf{W}_{k-1:1}$, show that

$$\dot{\mathbf{W}} = \frac{1}{L} \sum_{k=1}^L \mathbf{U}_L \mathbf{\Sigma}^{2L-2k} \mathbf{U}_L^\top \left(\mathbf{W}^* - \mathbf{U}_L \mathbf{\Sigma}^L \mathbf{V}_1^\top \right) \mathbf{V}_1 \mathbf{\Sigma}^{2k-2} \mathbf{V}_1^\top.$$

f) Alternatively, show that the gradient flow can be expressed solely in terms of $\mathbf{W}$:

$$\dot{\mathbf{W}} = \frac{1}{L} \sum_{k=1}^L [\mathbf{W} \mathbf{W}^\top]^{\frac{L-k}{L}} (\mathbf{W}^* - \mathbf{W}) [\mathbf{W}^\top \mathbf{W}]^{\frac{k-1}{L}}.$$

Problem 2 (Network near initialization):

Given an input vector $\mathbf{x} \in \mathbb{R}^d$, consider a shallow neural network defined by

$$f(\mathbf{W}) := \sum_{j=1}^m a_j \sigma(\mathbf{w}_j^\top \mathbf{x}), \quad \mathbf{W} := \begin{bmatrix} \leftarrow \mathbf{w}_1^\top \rightarrow \\ \vdots \\ \leftarrow \mathbf{w}_m^\top \rightarrow \end{bmatrix} \in \mathbb{R}^{m \times d},$$

where σ is an activation function, $\mathbf{a} \in \mathbb{R}^m$ is a fixed (not trainable) vector of weights initialized such that $a_j \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\left(\pm \frac{1}{\sqrt{m}}\right)$ ¹, and $\mathbf{W} \in \mathbb{R}^{m \times d}$ are trainable weights.

We will consider the linearization of the function f around some initial weights $\mathbf{W}_0$ defined by

$$f_0(\mathbf{W}) = f(\mathbf{W}_0) + \langle \nabla f(\mathbf{W}_0), \mathbf{W} - \mathbf{W}_0 \rangle_F, \quad (3)$$

where $\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{tr}(\mathbf{A}^\top \mathbf{B})$ is the Frobenius inner product.

a) Show that

$$\nabla f(\mathbf{W}) = \mathbf{D} \mathbf{a} \mathbf{x}^\top, \text{ where } \mathbf{D} = \text{diag}(\sigma'(\mathbf{w}_i^\top \mathbf{x})) = \begin{pmatrix} \sigma'(\mathbf{w}_1^\top \mathbf{x}) & & & \\ & \sigma'(\mathbf{w}_2^\top \mathbf{x}) & & \\ & & \ddots & \\ & & & \sigma'(\mathbf{w}_m^\top \mathbf{x}) \end{pmatrix}.$$

b) Show that

$$\langle \nabla f(\mathbf{W}_0), \mathbf{W} - \mathbf{W}_0 \rangle_F = \sum_{j=1}^m a_j \sigma'(\mathbf{w}_{0,j}^\top \mathbf{x}) (\mathbf{w}_j - \mathbf{w}_{0,j})^\top \mathbf{x},$$

where $\mathbf{W}_0 = \begin{bmatrix} \leftarrow \mathbf{w}_{0,1}^\top \rightarrow \\ \vdots \\ \leftarrow \mathbf{w}_{0,m}^\top \rightarrow \end{bmatrix}$ denotes the weight at initialization.

c) Now assume the activation function σ is β -smooth, which satisfies:

$$|\sigma(r) - \sigma(s) - \sigma'(s)(r - s)| \leq \frac{\beta(r - s)^2}{2}$$

for all $r, s \in \mathbb{R}$. Show that for any $\mathbf{W} \in \mathbb{R}^{m \times d}$,

$$|f(\mathbf{W}) - f_0(\mathbf{W})| \leq \frac{\beta}{2\sqrt{m}} \|\mathbf{W} - \mathbf{W}_0\|_F^2 \|\mathbf{x}\|^2.$$

d) What do you conclude about the role of over-parametrization?

¹Equivalently, one can define $f(\mathbf{W}) := \frac{1}{\sqrt{m}} \sum_{j=1}^m a_j \sigma(\mathbf{w}_j^\top \mathbf{x})$ with $a_j \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(\pm 1)$.